Fertility, Housing, and Population Dynamics

From a stationary closure to a dated transition

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The household mechanism stays; the population closure changes

Last time we emphasized:

- households choose births, tenure, house size, and saving over the lifecycle;
- children raise desired housing space, while down-payment and tenure frictions make that space costly;
- housing costs therefore affect both fertility and the timing of homeownership;
- parents' housing and wealth choices affect their children through transfers and bequests.

The missing aggregate link was what happens after those choices:

$$\text{household choices} \longrightarrow \text{births today} \longrightarrow \text{new households later},$$
$$\text{new households} \longrightarrow \text{housing demand and prices} \longrightarrow \text{new household choices}.$$

The update keeps the household model and adds this population–housing feedback.

The simplified economy has young and old households

At date t, there are Y_t young households and O_t old households.

A young household earns y and chooses consumption c, housing h, and children n:

$$u(c,h,n) = c + \alpha \log h + v \log n, \qquad c + ph + \left[q + a(p+\omega)\right] n = y.$$

- p is the current price of one unit of housing services.
- α is the value of adult housing; v is the value of children.
- Each child uses a units of housing and q units of other goods.
- ω is an extra housing-access cost associated with accommodating children.

Old households no longer choose fertility. Their housing demand is

$$h_O(p) = \frac{\bar{h}_O}{p},$$

where $\bar{h}_O$ measures their demand for housing services.

Housing costs jointly determine fertility and housing demand

The young household's interior choices are

$$n(p) = \frac{v}{q + a(p + \omega)}, \qquad h_Y(p) = \frac{\alpha}{p} + an(p).$$

Thus a higher p has two effects: it reduces adult housing demand and raises the housing cost of a child, reducing fertility.

Aggregate housing supply is

$$H^s(p) = \bar{H} p^{\varepsilon},$$

where $\bar{H}$ is the scale of housing supply and $\varepsilon \geq 0$ is its elasticity. At each date, p_t clears the market:

$$Y_t h_Y(p_t) + O_t h_O(p_t) = H^s(p_t).$$

This one price therefore links household fertility to aggregate cohort sizes.

A steady state keeps prices and cohort sizes constant

Let $\bar{n}$ denote the children per young household needed to reproduce the next young cohort. Then

$$Y_{t+1} = \frac{n(p_t)}{\bar{n}} Y_t, \qquad O_{t+1} = Y_t.$$

A steady state is a constant triple (Y^*, O^*, p^*). It therefore requires

$$n(p^*) = \bar{n}, \qquad Y^* = O^*.$$

The child decision pins down the steady-state housing cost,

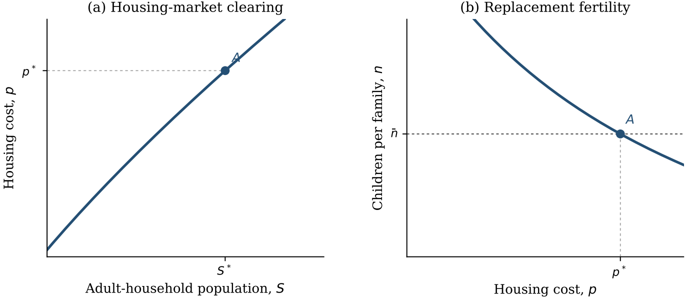
$$p^* = \frac{v/\bar{n} - q}{a} - \omega,$$

and housing clearing pins down the population scale $S^* = Y^* + O^*$:

$$S^* = \frac{2\bar{H}(p^*)^{1+\varepsilon}}{\alpha + \bar{h}_O + a\bar{n}p^*}.$$

Replacement determines the price; housing clearing determines how many households live at that price.

In steady state, population and housing cost are positively related



With equal young and old cohorts, the housing-clearing schedule is

$$S^H(p;v) = \frac{2\bar{H}p^{\varepsilon}}{h_Y(p;v) + h_O(p)}.$$

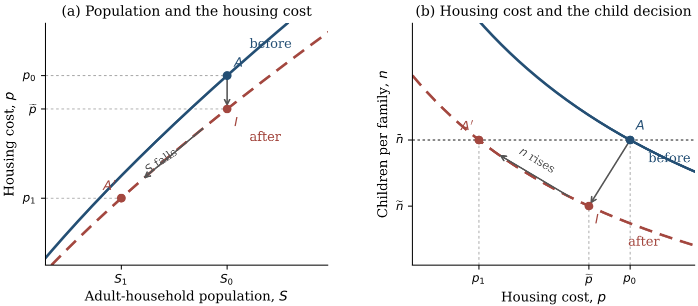
It slopes upward: supply weakly rises with p, while each household demands less housing.

The replacement condition in the right panel selects p^*; the left panel then selects S^*.

At a steady state, direct differentiation gives

$$\frac{dS^*}{dp} = \frac{2\bar{H}p^{\varepsilon}\left[(1+\varepsilon)(\alpha+\bar{h}_O) + \varepsilon a\bar{n}p\right]}{(\alpha+\bar{h}_O+a\bar{n}p)^2} > 0 \qquad \text{for } \varepsilon \geq 0.$$

A permanent fertility-preference decline has an impact and a transition



1. Impact. Let v fall permanently. Cohort sizes are inherited, but young households want fewer children and less space. The housing cost falls to $\widetilde{p}$, and fertility to $\widetilde{n} < \bar{n}$.

2. Transition. Fewer births create smaller future young cohorts. Population and housing demand fall, pushing the housing cost down further.

3. New steady state. The lower p_1 offsets part of the shock. Fertility returns to $\bar{n}$ at a smaller population $S_1 < S_0$.

Since $dp^*/dv = 1/(a\bar{n}) > 0$ and $dS^*/dp > 0$, a lower v implies both $p_1 < p_0$ and $S_1 < S_0$.

The arrows show impact and steady-state comparisons. The literal transition also depends on the evolving young-to-old cohort ratio.

Below-replacement fertility need not lower population immediately

Let $m_t = n(p_t)/\bar{n}$ be the replacement ratio and $\mu_t = O_t/Y_t$ the old-to-young cohort ratio. Population can rise with $m_t < 1$ whenever

$$\mu_t < m_t < 1.$$

A numerical example

$$100 \text{ young} + 70 \text{ old} \quad \longrightarrow \quad 80 \text{ young} + 100 \text{ old}$$

$m_t = 0.8 < 1$, but total households rise from 170 to 180.

Housing costs can rise as well if the aging cohort retains more housing than is released by the shrinking young cohort:

$$(1 - \mu_t)h_O(p_t) > (1 - m_t)h_Y(p_t).$$

U.S. interpretation: inherited cohorts can keep population and housing costs rising even after fertility falls below replacement.

The household model itself is unchanged

The household state x already contains wealth, income, location, tenure, house size, fertility history, and children at home. Households still solve the same sequential fertility, tenure, housing, saving, and moving problem.

What changes is the distribution. Let

$$g_t(j, x)$$

be the mass of age-j households in state x at calendar date t. In the old stationary exercise, this distribution was normalized back to a fixed population. In the transition, its total mass and age composition are allowed to evolve.

The household choices encoded in the transition operator $\mathcal{Q}_t$ determine where surviving households move across states:

$$\text{age, fertility, tenure, housing, and wealth at } t \quad \longrightarrow \quad \text{states at } t+1.$$

This preserves the intergenerational mechanism; it stops resetting the economy to the same cohort sizes.

Three equations add the aggregate feedback

1. **Birth-to-entry accounting.** Adjusted births b_t generate future entrant households after twenty years:
$$B_t = \frac{b_t}{2.1}.$$
The factor 2.1 combines sex composition, survival to entry age, and household formation.

2. **A calendar-time distribution law.** Existing households age according to their lifecycle choices, while the delayed birth cohort enters:
$$g_{t+1} = \mathcal{Q}_t g_t + \text{entrants}_{t+1}.$$

3. **Housing clearing at every date.**
$$H^s(P_t) = \int h_t(j,x)\, dg_t(j,x).$$

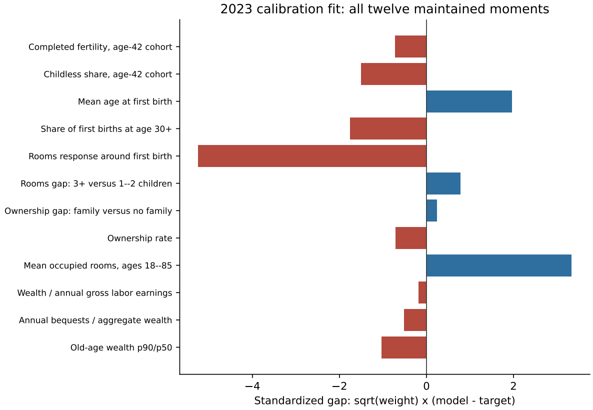
Sequential fertility, tenure, saving, and bequests are unchanged; only cohort masses now evolve over calendar time.

We calibrate an economy moving from 2007 to 2023

1. **Construct an old reference economy.** Choose the initial fertility preference so completed fertility is 2.1. This is a replacement normalization, not a measured 2007 target.
2. **Construct the 2007 state.** Reweight the old distribution only by observed 2007 age shares, preserving wealth, tenure, and fertility states within each age.
3. **Run the economy to 2023.** Feed in Census household totals and ACS age shares. Let the fertility preference decline linearly, and estimate the household parameters and total decline using twelve maintained moments, evaluated mainly in 2023.
4. **Check the path.** Compare untargeted model movements with period fertility, birth timing, ownership, rooms, house prices, and rents.
5. **Continue after 2023.** Hold preferences fixed and let the inherited household distribution and birth pipeline evolve.

The population and age bridge are inputs, not successes of the model. The 2007 economy is a dated initial condition, not literally the old steady state.

The 2023 fit has two important housing-size misses



Source: final dated-transition calibration; selected candidate 7.

Calibration facts

Loss: 50.742**.** Ten parameters and twelve moments.

Two exact independent repeats.

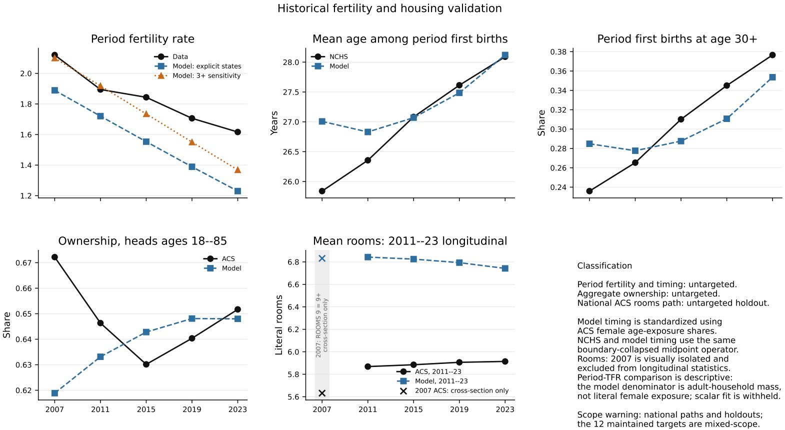
Main misses

- First-birth rooms: 0.273 versus 0.720
- Mean rooms: 6.742 versus 5.780

They contribute 38.57 of the total loss.

The PSID rooms target is provisional because its prepath is not flat.

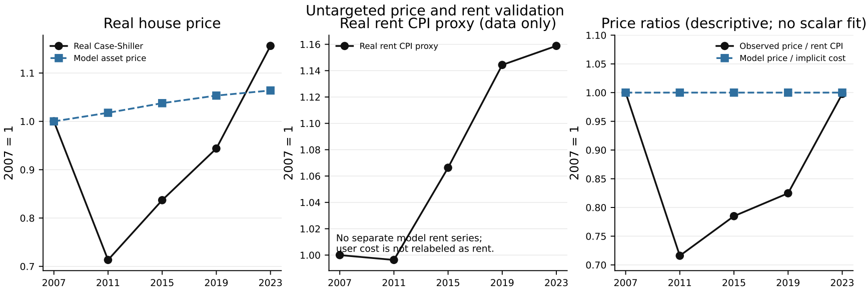
Historical validation is mixed rather than hidden



Reading the figure: timing is close by 2023; the model misses the ownership cycle and rooms trend, and its fertility decline is too steep.

National series are holdouts; maintained targets use mixed samples and vintages.

The model does not reproduce the house-price cycle



- The model cannot reproduce the 2007–2011 house-price collapse and later rebound.
- With temporary-equilibrium expectations and a fixed user-cost relation, it has no independent price-to-rent dynamics.

The exercise should therefore be read as a demographic-housing transition, not as a complete account of the post-2007 housing cycle.

Observed house prices: real Case-Shiller. The rent CPI is shown only as a descriptive proxy, not as the model's user cost.

What exactly is the post-2023 continuation?

Both exercises begin from the **same calibrated 2023 distribution**: the same ages, wealth, tenure, fertility histories, children at home, and birth pipeline. We then hold the 2023 fertility preference fixed and introduce no policy.

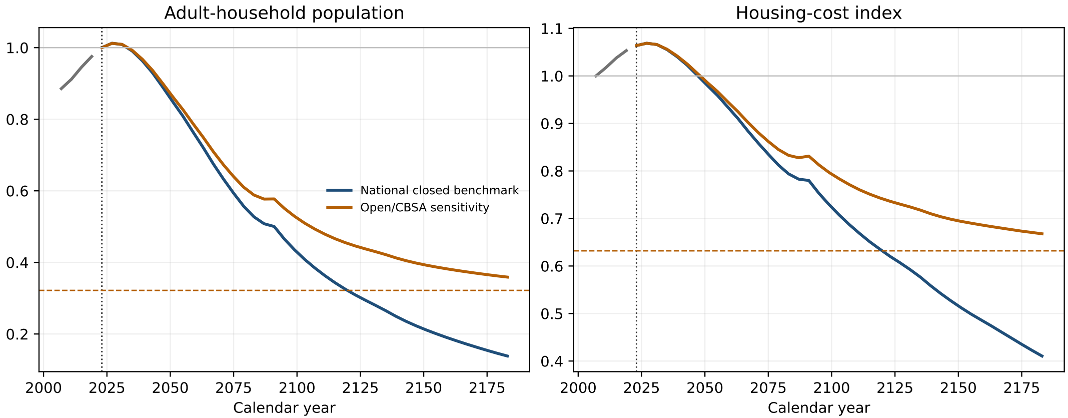
The two paths differ only in how new households enter after 2023:

- **Closed benchmark.** There is no outside entry. Future entrants come only from earlier birth cohorts.
- **Open sensitivity.** A fixed outside flow M and a fixed local-retention rate ρ continue from the old reference economy.

Because births take twenty years to become entrant households, the first post-2023 years are largely predetermined. Inherited cohorts can make population and housing costs rise briefly even though reproduction is below replacement.

These are controlled equilibrium experiments, not forecasts. The open case does not hold the outside-origin entrant *share* fixed: 0.169 only normalizes the old reference state.

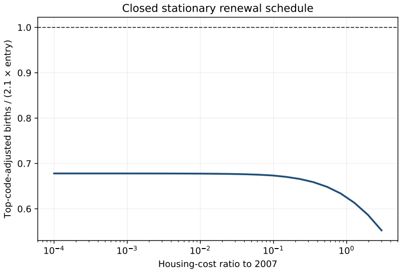
Demographic momentum delays, but does not prevent, decline



Both paths peak slightly in 2027 because the initial age structure and birth pipeline are inherited. Afterward, the closed economy contracts much faster. By 2183, population is 0.138 of its 2023 level in the closed benchmark and 0.359 in the open sensitivity; asset prices are 0.386 and 0.628.

In the open stationary endpoint, the realized outside-origin entrant share is 0.465 because the fixed flow M becomes large relative to a shrinking population.

The fitted closed economy has no positive terminal steady state



25-point grid, $P/P_0 \in [10^{-4}, 3]$; not a global theorem.

At each stationary price P, let $E(P)$ be entrants and $B(P)$ the future households generated by births. A positive closed steady state requires $B(P)/E(P) = 1$.

Instead, over the audited grid,

$$0.552 \leq B(P)/E(P) \leq 0.678 < 1.$$

Even almost free housing does not reproduce the entrant cohort.

Therefore: the closed continuation is a finite-horizon experiment, not a transition between two positive steady states.

Where this leaves the paper

- **The theoretical contribution is compact.** The existing housing–fertility mechanism is embedded in a population closure. Lower fertility eventually reduces population and housing costs, while inherited cohorts can make both rise initially.
- **The quantitative exercise is a dated transition.** We initialize in 2007, fit the maintained 2023 moments, and keep imposed demographic inputs separate from historical holdouts.
- **The present fit is informative but incomplete.** Several 2023 moments fit well, but housing quantities, the ownership cycle, and the house-price path do not.
- **The equilibrium problem is substantive.** The current fertility response is too weak to restore replacement at any audited positive price, so the closed continuation has no positive terminal steady state.

The immediate decision is narrow: either present a finite closed benchmark with an explicit open sensitivity, or re-estimate the housing-cost-to-fertility slope under a closed-root requirement before claiming a transition between steady states.

Policy analysis comes only after this equilibrium choice is settled.